

Time-independent Schrodinger equation

Consider the Schrodinger equation

$$i\hbar \frac{\partial}{\partial t} \Psi(x, t) = H \Psi(x, t)$$

Let $\Psi(x, t)$ be separable as

$$\Psi(x, t) = \sum_k \varphi_k(t) \phi_k(x)$$

Let it be the case that for all k

$$i\hbar \frac{d}{dt} \varphi_k(t) \phi_k(x) = E_k \varphi_k(t) \phi_k(x)$$

where all E_k are constant. Then by the Schrodinger equation we have

$$E_k \varphi_k(t) \phi_k(x) = H[\varphi_k(t) \phi_k(x)]$$

If H is independent of t then $\varphi_k(t)$ cancels and we have

$$E_k \phi_k(x) = H \phi_k(x)$$

Hence $\sum_k \phi_k(x)$ solves the time-independent Schrodinger equation

$$\sum_k E_k \phi_k(x) = H \sum_k \phi_k(x)$$

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